Lance Matthew C. Jallorina BSIT 2-1 3/19/2026 Midterm Seatwork #3 - Basic ATM System

```
import java.util.Scanner
public class ATMTransactionCounterJallorina {
    public static void main(String[] args){
        int balance = 8000;
        int transactionscounter = 0;
        boolean active = true;
        Scanner input = new Scanner(System.in);
        while (active) {
            System.out.println("===== ATM MENU =====");
            System.out.println("1. Check Balance");
            System.out.println("2. Deposit");
            System.out.println("3. Withdraw");
            System.out.println("4. Exit");
            System.out.print("Enter Choice");
            int option = input.nextInt();
            switch (option) {
                case 1:
                    transactionscounter++;
                    System.out.println("Balance: P" + balance");
                    break;
                case 2:
                    System.out.print("Deposit amount: P");
                    int addbal = input.nextInt();
                    balance += addbal;
                    System.out.println("Deposited: P" + addbal");
                    System.out.println("New Balance: P" + balance);
                    break;
                case 3:
                    System.out.print("withdraw amount: P");
                    int withdraw = input.nextInt();
                    transactionscounter++;
```

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```
                    if (withdraw > 2000) {
                        System.out.println("Exceeds maximum withdrawal of ₱ 2000!");
                    } else if (withdraw > balance) {
                        System.out.println("Insufficient balance!");
                    } else {
                        balance -= withdraw;
                        System.out.println("Withdrawn: ₱" + withdraw);
                        System.out.println("New Balance: ₱" + balance);
                    }
                    break;
                case 4:
                    System.out.println("Exiting.... Goodbye!");
                    active = false;
                    break;
                default:
                    System.out.println("Invalid choice. Try again");
                }
            }
            System.out.println("Transaction Count: " + transactionsCounter);
            System.out.println("Final balance: ₱" + balance);
        }
    }
```

Short Exploration:

How the while loop works: the "while(active)" loop keeps the ATM menu running continuously until the user selects option 4, which sets "active = false" and stops the loop. Every time a valid menu option is selected, the loop repeats and displays the menu again.

How the switch-case controls the menu: the switch (option) acts as the decision maker that directs the program flow based on the integer value of the option. Each case represents a specific function, such as checking the balance, depositing money, or withdrawing funds. The break statement of each case is crucial because it prevents executing the subsequent cases after a case is executed.

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How the if-else validates the withdrawal: The if-else logic within the withdrawal serves as the security gate to ensure all transaction rules are met before updating the balance. It first evaluates if the withdrawal amount exceeds the ₱2000 limit, then checks if the user actually has enough money in their account. If both conditions pass, the else block executes the withdrawal, or an error message is displayed and the balance remains the same.